Angel Vargas

Automotive Systems Engineer

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Soft Skills

Assertive **Communication**

Innovative **Thinking**

Self-Management **& Accountability**

Leadership & **Coaching**

Resilience & **Versatility**

Problem Solving & **Decision Making**

Continuous **Learning**

Hard Skills

Project **Management**

Requirement **Engineering**

Vehicle Network Architecture **Diagnostic**

Digital **Transformation**

Machine **Learning**

Quality Management **Systems**

Testing and **Validation**

Product Lifecycle **Management**

Systems **Engineering**

Languages

English (CEFR C1)

French (CEFR A1)

German (CEFR A1)

Spanish (Native)

Profile

Innovation-driven Automotive Systems Engineer who has carried breakthrough features **from concept to launch** for global pickup and SUV lines. Excelling in **feature design, diagnostics, and troubleshooting**, pairing my technical expertise with a character grounded in resilience, creativity, and integrity.

Skilled at uniting hardware, software, and validation, I **lead multidisciplinary teams** from voice of the customer through vehicle sign-off, **delivering high-performance systems** on schedule. Driven by **emerging technology,** I focus on accelerating the transition to next-generation high-performance and autonomous vehicles.

Professional Experience

Diagnostics Design Engineer - Ford Motor Company, GTBC Mx

JULY 2025 – PRESENT

- Spearheaded the "End-to-End ECU Diagnostics Robustness" project; analyzed systemic pain points across ECU design, delivering a **new standardized design framework.** Resulting for the pilot iteration in **150+ diagnostic requirements** and test cases for pilot features by synthesizing DFMEAs and Ford/Supplier ECU specifications.
- Targeted **high-warranty "Noisy DTCs"** for TPMS, Key Fob, and VIN Authentication; **authored 23+ requirements** to eliminate false-positive triggers and reduce future service costs.
- **Bridge cross-functional teams** (Architecture, SW, Module D&Rs) to close diagnostic coverage gaps and audit ECU documentation for next-gen vehicle architectures.
- Designing next-gen "Enhanced Data Recording" methods (in-development) to optimize engineering troubleshooting.

Feature Systems Engineer - Ford Motor Company, GTBC Mx

SEPTEMBER 2021 – JULY 2025

- Served as both Core (feature concept & design) and Application (vehicle integration/validation) engineer the **Power Running Boards** (PRB), **Onboard Scales & Smart Hitch** (OBS/SH), and **Automatic Vehicle Shutdown** (AVS) guiding each from early concept to global launch.
- Authored **≈700 layered requirements** (concept, logical and technical) and **400+ test cases**, achieving full traceability and closing legacy gaps.
- Launched NVH lab test for PRB —**cutting noise level 36 %**—and deleted a low-use hardware, **saving ≈$2 M** per model year.
- Built Feature usage and **Failure-Mode-Avoidance dashboards** that analyzed in-use data and DFMEA failure modes; insights justified OBS/SH content deletion on low-spec F-Series (25 MY) and reinforced PRB **cost-down actions**.
- Calibration refinements and targeted diagnostics **cut PRB warranty spend by ≈$1.4 M** (23MY baseline); delivered all three features with zero open critical issues at launch.
- Filed **12 innovation disclosures** and mentored six onboard engineers to the systems engineering and validation organization.
- Managed US Plant rotations (DTP/KTP) ensuring **98%+ first-time-through** build quality for Job #1.

Quality Management Engineer - BMW Group, Plant San Luis Potosí

JULY 2020 – JUNE 2021

- Supported **G42 Conformity-of-Production** and ISO 9001 audits—both passed with zero major findings.
- Deployed Power BI/Power Apps suite for the Quality Management System, **cutting issue-closure time by 30%.**
- Launched a “Quality Mindset” campaign that trained 120+ process leaders and boosted **proactive defect containment actions 25 %** in six months.

Tools & Software

Microsoft 365 Suite (Advanced)
Atlassian Platform – Jira (Advanced)
CANalyzer, CAPL (Advanced)
Diag Engineering Tool (Advanced)
Project Management – Agile | Scrum (Advanced)
MagicDraw & JAMA (Intermediate)
ISO 9001, IATF 16949 (Intermediate)
Quality Core Tools (Intermediate)
Catia V5, SolidWorks & NX (Intermediate)
VS Code – Python, C++(Intermediate)
Power BI & Tableau (Intermediate)
Siemens TestLab (Basic)
Minitab (Basic)

Professional Cert

ID - 12898975

Location

México - Open to relocation

Contact

Activities & Interests

Sports Club Member – Tennis Player and Part of the Swimming Club.

Cooking and Bakery - Following family traditions and likes on desserts.

Travel Enthusiast – Extensive travel experience fostering adaptability and cultural awareness.

Education

Bachelors in Automotive Systems Engineering at Instituto Politécnico Nacional, U.P. ESIME Zacatenco, México City, México
AUGUST 2017 – FEBRUARY 2021

Specialty in Engine and Thermo-fluids Design at Instituto Politécnico Nacional, U.P. ESIME Ticomán, México City, México
AUGUST 2019 – FEBRUARY 2021

Courses & Certifications

"Six Sigma Green Belt Certification", Ford Motor Company Mexico
MAY 2023

"Feature/Platform Systems Engineering", Ford Motor Company Mexico
NOVEMBER 2023

"VECTOR CAPL Initialization", Vector NA, Inc.
OCTOBER 2024

"Computer Programming Literacy", Ford Motor Company
DECEMBER 2024

"AUTOSAR Architecture", Udemy Business
JULY 2025

"Machine Learning A-Z: AI + Python", Udemy Business
AUGUST 2025

"Google Project Management Professional Certificate",
Coursera-Google Career Certificates
JANUARY 2026

"EF SET English Test", EF SET
JANUARY 2026

International Experience

International Assignment, Ford Motor Company, Michigan and Kentucky, USA
4-MONTHS LAUNCH ROTATION 2022-2025

- Resolved critical launch issues on-site, keeping PRB, OBS/SH, and AVS features on schedule and preserving **≥98 % first-time-through** build quality.
- Optimized feature performance in extreme temperatures (−40 °C to +50) °C thermal chambers and anechoic labs, improving cycle time, obstacle detection, and interior noise.
- Devised cross-platform test methods—from breadboard to full vehicle—later adopted as the template for next-gen truck and SUV programs.
- Led diagnostics triage with U.S. module, software, and supplier teams; actions cut projected warranty spend and fed **25+ lessons learned** into Ford's feature specifications.